

JANUARY, 2019

Global nickel short-term outlook January 2019



Executive summary

Nickel prices started 2019 at their lowest level since September 2017. However, new long positions spurred a recovery that was further supported by a short squeeze mid-month and then, somewhat unconvincingly, by concerns about the risk posed to Vale's nickel operations by fallout related to the tailings dam collapse at the company's Corrego do Feijao iron ore mine in Brazil.

Since our December report, the principal revisions to our balances are dominated by additional supply from Chinese and Indonesian NPI sources, as well as the new Indonesian hydrometallurgical plants. As these are only partly compensated for by increases in demand, the most notable being a second stainless mill in Indonesia from next year, the result is a reduction in our projected deficits over the next three years. Crucially, the market will remain in deficit but in 2019, a deficit of only 28 kt is close enough to balance to undermine the case for higher prices. The same is true of 2021 and our revised forecasts reflect this less bullish outlook.

Key forecasts

	2018		2019		2020		2021	
Global (kt)								
Mine Production	2282	+4.5%	2455	+7.6%	2587	+5.4%	2727	+5.4%
Smelter Production	2192	+6.0%	2295	+4.7%	2385	+3.9%	2476	+3.8%
Refined Production	2243	+5.5%	2360	+5.2%	2466	+4.5%	2582	+4.7%
Consumption	2303	+5.2%	2388	+3.7%	2569	+7.6%	2607	+1.5%
Market Balance	-59		-28		-103		-25	
Cash LME price								
(\$/t - Current)	13114		12119		15331		14339	
(c/lb - Current)	595		550		695		650	

Source: Wood Mackenzie

Supply-demand balances

Nickel prices rise in January Nickel prices dropped below US\$11,000/t (US\$5.00/lb) in December 2018 and moved towards US\$10,000/t (US\$4.50/lb) so that they started 2019 at US\$10,437/t (US\$4.73/lb, cash), their lowest level since September 2017. However, new long positions supported a recovery to US\$11,400/t (US\$5.17/lb) by 11 January 2019. Prices continued to rise supported by short covering on the 14-15th, triggered by a short squeeze that resulted in temporary backwardation, and finally reached US\$11,537/t (US\$5.40/lb) at the close on 24 January.

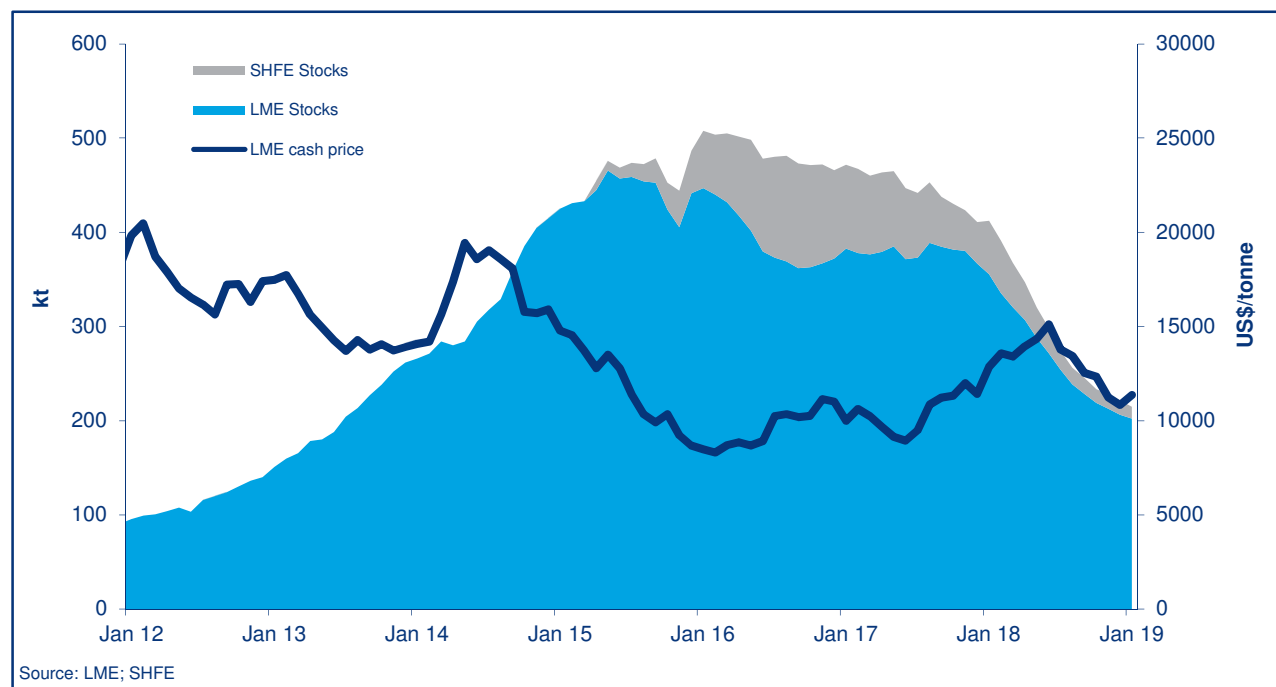
Vale concerns push prices higher Despite there being no obvious fundamental reason, prices surged again from 25 January, reaching intra-day highs well above US\$12,000/t (US\$5.45/lb) on the 30th. The cause appears to be concerns about the risk posed to Vale's nickel operations in the aftermath of the tailings dam collapse at Corrego do Feijao iron ore mine in Brazil. Clearly, as the mine extracts iron ore, the disaster has no direct influence on nickel supply. However, the clean up bill will be large and as this is Vale's second such disaster in four years (after Samarco in 2015), safety measures at its other operations could come under close scrutiny. The overall cost to Vale is likely to be substantial, so it is conceivable that this might have a bearing on the company's nickel business. But even if that is possible, the impact is months, if not years, away.

Exchange stocks still falling Exchange nickel stocks continued to decrease from 222 kt at the end of 2018 to 215 kt in the final week of January 2019. SHFE inventory dropped from 15.3 kt in December to 12.7 kt on 25 January. In contrast, LME



stocks declined by nearly 10 kt, to 198 kt on 14 January, only to then start increasing again, returning to 203 kt on 28 January. This pattern is consistent with the aforementioned short squeeze that was played out by traders on the 14-15 January.

LME and SHFE nickel stocks, and nickel prices



Movement of Chinese premiums varies In China, bonded warehouse inventory continued to fall, nearing 26 kt in mid January compared with around 60 kt in the middle of 2018. This is consistent with the increase in nickel full plate CIF premium, from US\$190-200/t in late December to around US\$210-220/t in late January. Despite a closed arbitrage window, full plate cathode transaction volumes surged in the second and third week of the month. This was probably due to the potential VAT reduction from 16% to 14%, meaning there could be a potential 2% tax saving for traders purchasing now and selling later. Briquette premiums, on the other hand, were unchanged at US\$250-260/t due to the low season and upcoming Chinese new year. Discounts on FeNi were also broadly unchanged. As there is sufficient FeNi supply and worries about a rise in NPI production in 2019, the downward pressure on the FeNi price remains. For a FeNi containing 32% Ni, we estimate a discount of around US\$300/t (2-3%) in Q1 2019.

Projected deficits getting smaller The revisions we have made to our balances since December are dominated by additional supply from Chinese and Indonesian NPI sources, as well as the new Indonesia hydrometallurgical plants. As these are only partly compensated for by an increase in demand, the result is a reduction in the projected deficits over the next three years. Crucially, the market will remain in deficit through to 2021, the cumulative shortfall since the end of 2015 reaching 320 kt by then. But in 2019, a deficit of only 28 kt is close enough to balance to undermine the case for higher prices. The same is true of 2021 and our price forecasts have been revised to echo this less bullish outlook.

Forecast prices revised down in 2019 This month's rally in the nickel price warranted a slight increase to our forecast of the Q1 2019 average price. However, this is more than offset by a reduction in forecast for the rest of the year because of increasing supply and a narrowing deficit. Indeed, we expect a small surplus in Q3 2019. Added to Brexit uncertainty having at least a temporarily negative impact on prices, we expect the nickel price to fall back to sub-US\$11,000/t (US\$5.00/lb) values in March-April before starting to climb. That rally will stabilise in Q3 before picking up again as the market looks ahead to a more substantial deficit in 2020. On that basis, the forecast average annual price for 2019 now stands at US\$12,120/t (US\$5.50/lb).



Global nickel quarterly supply-demand balance

	Refined Supplies	Refined Consumption	Balance	CashPrice		Stocks	
				\$/tonne	c/lb	kt	Days of Cons
2018Q1	549	567	-18	13283	603	1386	220
Q2	560	579	-20	14463	656	1366	214
Q3	552	566	-14	13240	601	1352	220
Q4	582	590	-8	11469	520	1344	210
Total/Avg	2243	2303	-59	13114	595		
2019Q1	572	573	-1	11358	515	1344	211
Q2	591	598	-7	11611	527	1337	203
Q3	601	595	6	12162	552	1343	208
Q4	596	622	-27	13345	605	1316	195
Total/Avg	2360	2388	-28	12119	550		
2020Q1	617	628	-12	14367	652	1305	187
Q2	617	649	-31	15873	720	1274	179
Q3	608	639	-32	15579	707	1242	179
Q4	625	653	-28	15506	703	1214	171
Total/Avg	2466	2569	-103	15331	695		
2021Q1	643	632	11	14183	643	1224	174
Q2	645	665	-20	14661	665	1204	165
Q3	646	649	-2	14404	653	1202	170
Q4	648	661	-13	14110	640	1189	165
Total/Avg	2582	2607	-25	14339	650		

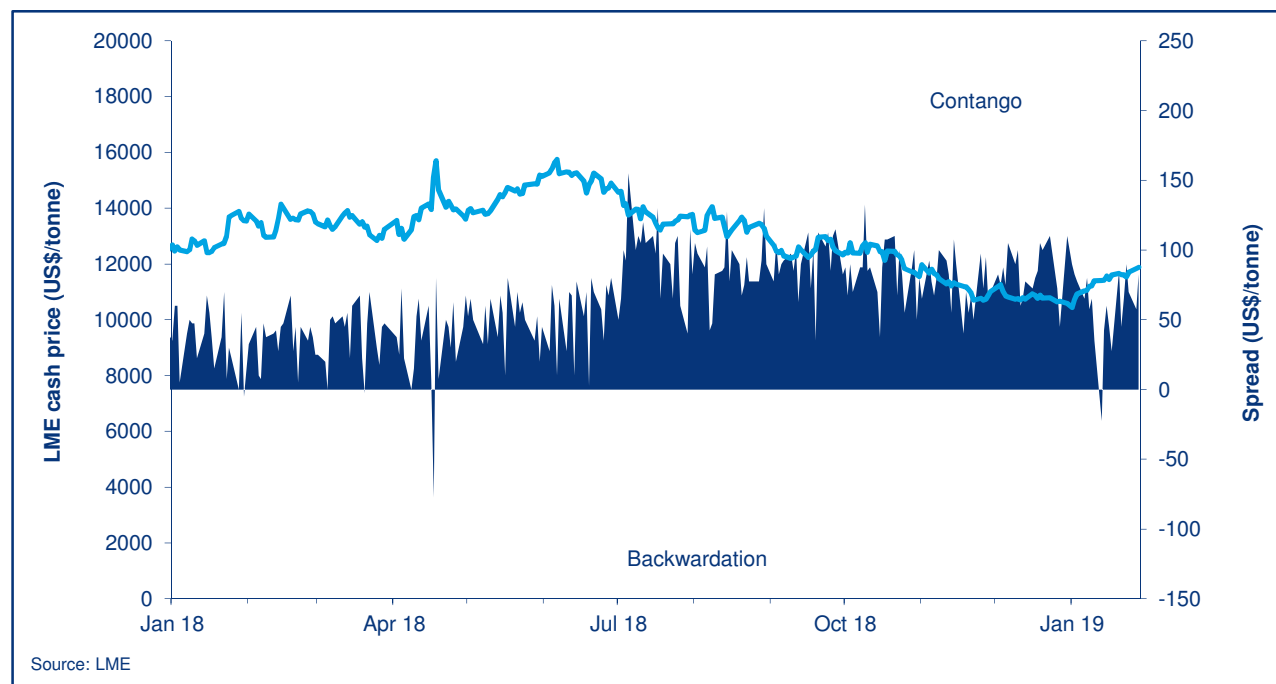
Source: Wood Mackenzie

Prices target US\$16,000/t in 2020 Expansions at two large Chinese stainless mills as well as start up at greenfield capacity contribute towards strong growth in nickel demand during 2020. This ultimately leads to a 100 kt deficit that will be boosted further by ramp up at a second stainless mill in Indonesia. For that reason we expect the price improvement that begins in late 2019 will accelerate in 2020, peaking at around US\$16,000/t (US\$7.25/lb). This will give an average price of around US\$15,330/t (US\$6.95/lb).

Price retreat in 2021 Retreat will start before the end of the year as the more balanced market of 2021 comes into view. New hydrometallurgical capacity starts to produce nickel chemicals for the battery segment in that year. This will exert some downward pressure on prices that we believe will steady within a range of US\$13,000-15,000/t (US\$5.90-6.80/lb), to give an average for the year of US\$14,340/t (US\$6.50/lb).



LME cash and 3-month spread



Demand

Revisions focused on China and Indonesia This year has started in much the same vein as 2018 finished, with our latest revisions centred on China and Indonesia. Net changes amount to a cut in world nickel use of up to 33 kt a year in 2018 and 2019, countered by increases of 17 kt and 48 kt in 2020 and 2021, respectively. The near-term cuts relate to increased stainless scrap use by Tsingshan in China and Indonesia, and cuts in melting in Taiwan. In contrast, the later increases are mostly linked to ramp up of a second stainless mill in Indonesia.

World nickel demand up 4% a year to 2021 We estimate that world stainless melt production increased by 5% in 2018, to 50.7 Mt. Growth will slow in 2019 to less than 2%, total output reaching to 51.5 Mt, but will surge again in 2020, by 8%, to 55.7 Mt, due to new capacity in China and Indonesia. Growth will slow again in 2021. Nickel demand also increased by around 5% in 2018, to 2.3 Mt. Growth over the next three years will mimic that of stainless, slowing to 3.7% in 2019, to 2.39 Mt before escalating in 2020, to 2.57 Mt, and decelerating in 2021, to 2.61 Mt.

Quarterly global nickel consumption in stainless steel (kt)

	2018	2019					2020				
	Yr	Q1	Q2	Q3	Q4	Yr	Q1	Q2	Q3	Q4	Yr
SS output	50675	12264	13005	12837	13436	51543	13615	14184	13754	14099	55653
Aust output	39311	9389	9971	9879	10341	39580	10402	10850	10509	10791	42552
Scrap (%)	40.3	40.4	39.8	38.4	38.5	39.3	39.4	39.0	37.9	38.3	38.6
Aust (%)	77.6	76.6	76.7	77.0	77.0	76.8	76.4	76.5	76.4	76.5	76.5
Primary Ni	1576	389	412	409	430	1639	432	451	443	452	1777

Source: Wood Mackenzie



Quarterly refined nickel consumption by first use (kt)

	2018	2019					2020				
	Yr	Q1	Q2	Q3	Q4	Yr	Q1	Q2	Q3	Q4	Yr
TOTAL	2303	573	598	595	622	2388	628	649	639	653	2569
Stainless Steel	1576	389	412	409	430	1639	432	451	443	452	1777
Non-Stainless	726	184	186	186	193	749	197	198	196	201	792
Alloy Steel	89	22	22	22	23	89	23	23	23	23	93
N-F Alloys	236	59	59	59	61	238	62	62	61	62	247
Plating	137	35	35	35	35	139	36	35	35	36	142
EV Batteries & ES	96	29	30	32	34	125	36	37	38	39	149
Foundry	64	16	17	16	17	66	17	17	16	17	67
Other	103	23	23	23	23	92	24	24	23	23	94

Source: Wood Mackenzie

Southeast Asia

Indonesia

Tsingshan Indonesia will go to 3 Mtpa In 2018, Tsingshan was unable to fully utilise its 3 Mtpa Indonesian stainless mill. There were two prime reasons for this: its stainless shipments to China had a detrimental impact on the Chinese market; and offtake arrangements with buyers elsewhere in Asia took time to organise. A third factor was the finite availability of nickel units within the Tsingshan group; essentially the existing 20 RKEF lines producing NPI in Morowali are not sufficient to support 3 Mtpa of austenitic stainless production. The addition of four more lines this year (and two in 2020) should ease this problem. We also expect the expansion of Tsingshan's export business to continue in 2019, and this too will strengthen the case for commercial start up of the third 1 Mtpa line.

New Indian rolling mill will facilitate expansion A key facilitator in this could be Tsingshan's own 600 ktpa stainless cold rolling mill in India, which is expected to start up in the first half of 2019. As our current outlook assumes that line #3 in Indonesia starts in July, output from it in 2019 would be limited to a maximum of around 500 kt, all of which could all be absorbed by the new Indian rolling mill. Thus, the impact of Tsingshan's Indonesian expansion in 2019, might only be felt in India rather than across Asia. That prospect is a matter of concern for some Indian producers (see our January 2019 Insight on the evolution of India's stainless sector), especially given the reaction to Indonesian imports by Taiwanese mills (see Taiwan, below).

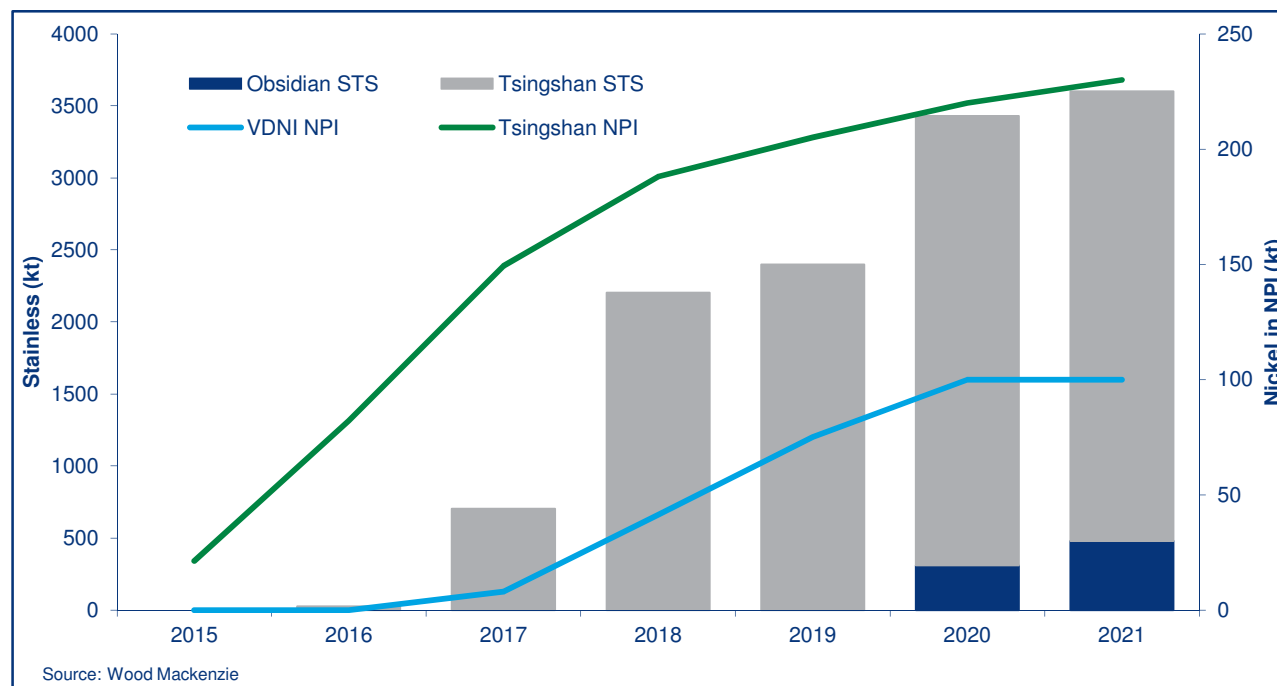
Delong building Indonesian stainless mill Given the impact that Tsingshan's Indonesian stainless mill is having on competing SE Asian stainless steelmakers, it will be interesting to see if Jiangsu Delong's Indonesian stainless venture makes a similar impression. PT Obsidian Stainless Steel, a joint venture between Delong and Chinese state-owned Xiamen Xiangyu Logistics Group, was originally scheduled to begin operations in Q2 2019. However, our understanding of progress to date indicates that such a timeline is now optimistic, certainly for the entire 2.5 Mtpa facility that is planned.

Obsidian will take VDNI nickel from 2020 Anecdotal sources indicate that the first stage of the melt shop may only be 600 ktpa, the equivalent to one of the six 100t AODs purchased by Delong in August 2018. As Delong's NPI facility – Virtue Dragon Nickel Industry (VDNI) - is based on the same site, most of the infrastructure required for this first stage stainless mill is probably already in place. If so, and allowing for around 15 months construction period, it is possible that first production could take place late in the year, or more likely early 2020. Furthermore, we expect the ramp up of NPI production at VDNI to 100 kt contained nickel in 2020 (cf, 41 kt in 2018) to provide the nickel units to feed this stainless mill.

Indonesian stainless to reach 3.6 Mt in 2021 Consequently, we have included Obsidian (stage one only) in our base case for the first time, tentatively from Q2 2020. With Tsingshan Indonesia and Obsidian both in production, Indonesian stainless melt output could increase to 3.6 Mt in 2021, thereby raising primary nickel uptake to 254 kt, compared with 156 kt in 2018.



Indonesian ramp up in NPI to feed ramp up in stainless steel



Taiwan

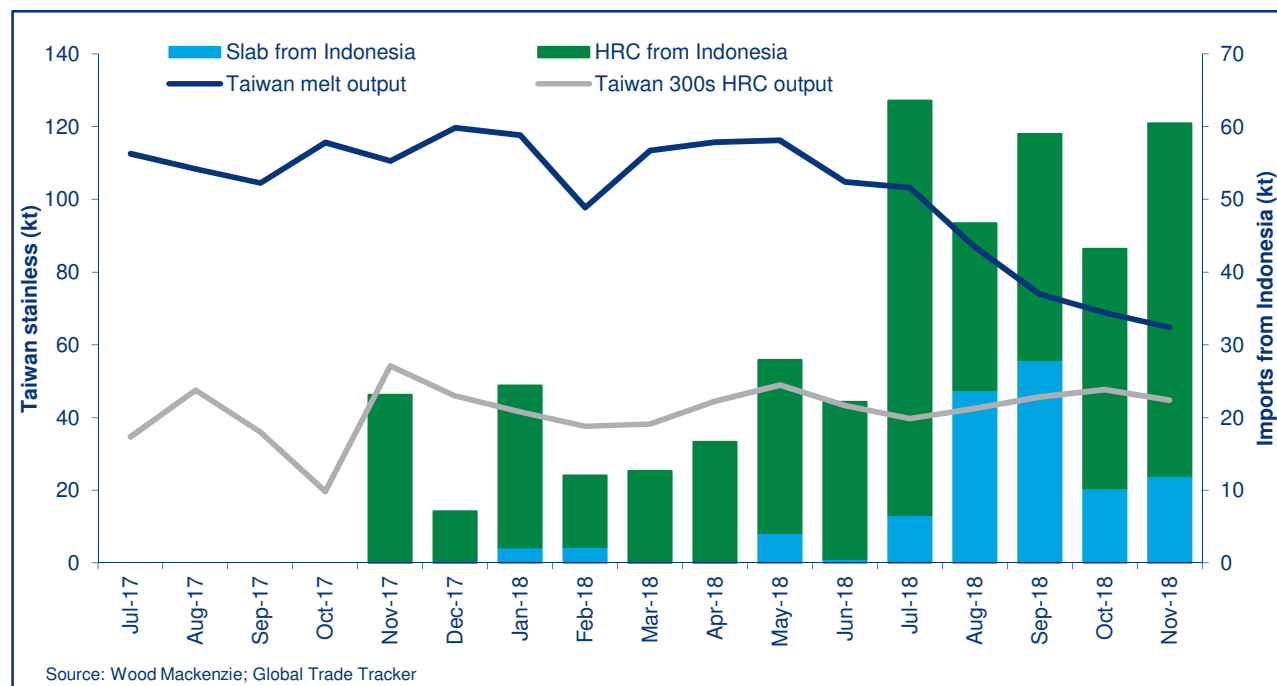
Taiwan imports of stainless slab and HR surge Up to now the surge in Tsingshan's exports of slab and hot rolled (HR) stainless to Taiwan has triggered a dramatic reduction in Taiwanese melt shop activity. In the first half of 2018, average Taiwanese output of stainless melt was 111 kt a month but this dropped to 60-70 kt a month in Q4. Imports of slab from Indonesia increased from 2 kt a month in H1 2018 to 20 kt a month in August to October, whereas imports of HR stainless increased from 30 kt in 2017 to 300 kt in January to November 2018. The net result is that the Taiwanese mills can maintain cold-rolled (CR) product output while reducing activities in their more cost intensive melt shops and HR lines.

Taiwanese stainless melt output cut Interestingly, there is broad parity between the increase in HR imports from Indonesia and the decrease in HR imports from China. Thus, the cut in Taiwan's melt shop production appears to be primarily influenced by volumes of imported Indonesian slab, which at the end of 2018 stood at around 20 kt a month. We have used that as a guide to the cut in melt output that might be possible in 2019 and reduced our forecast by 160-200 kt a year in 2019 and 2020.

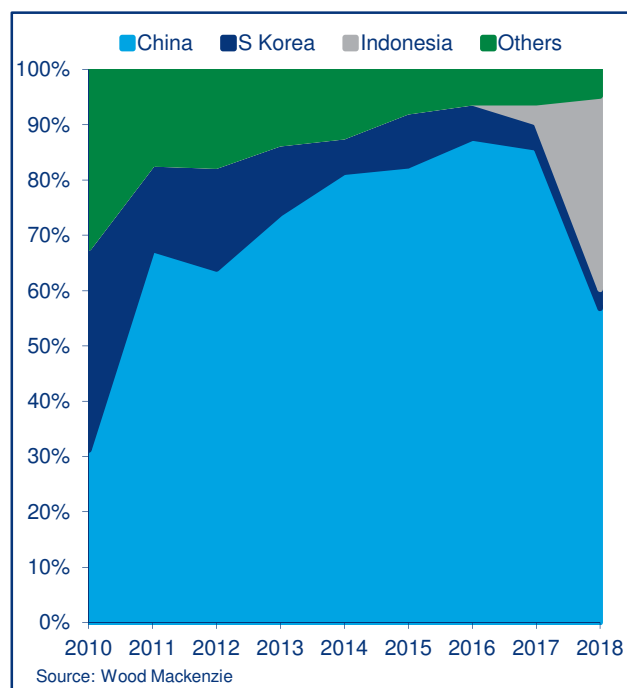
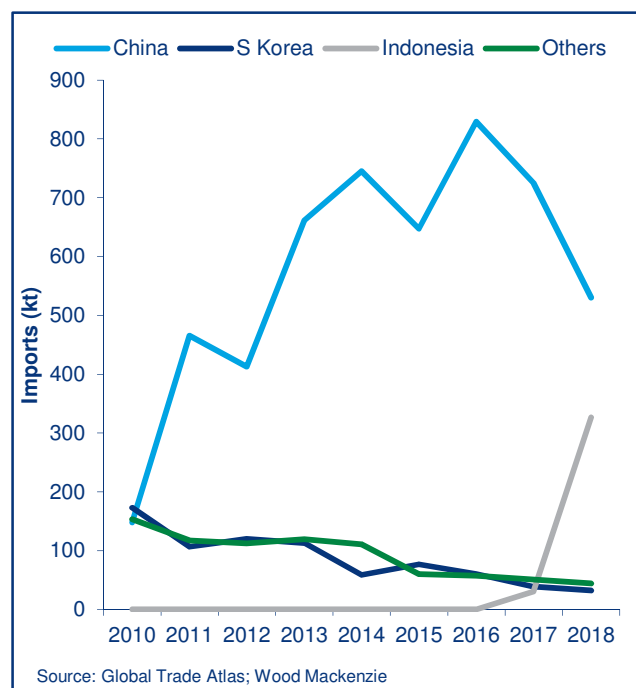
But Taiwanese nickel uptake unchanged At the same time we have adjusted austenitic ratios to allow for these growing tonnages of imported slab. Our calculated ratios are based on the HR stainless product mix as reported by the TSIIA which since mid-2018 has included HR product made from imported slab, so this component has to be removed from our calculation. Even after this adjustment is made, the Taiwanese austenitic ratio has still increased sharply, from 78-80% through 2017 and H1 2018 to 87.7% in Q4 2018 (89% if imported slab is included). In nickel usage terms, the forward projection of such higher values in 2019 and 2020 counters any cut in use through lower melting rates.



Taiwan's stainless melt output falling due to rising imports of Indonesian slab



Indonesian HR stainless replacing that from China





Japan

Japanese stainless output aiming for its highest in four years The Japanese stainless steel sector is on its way to enjoying its best year since 2014, in terms of stainless melt production growth at least. Admittedly, December's data will have to be strong if it is to match our estimated 3.30 Mt for the year (compared to 3.33 Mt in 2014), but as this has been the pattern of recent years we are confident that it will be close. Supporting that confidence are lower than average stainless product stocks, which stood at 130 kt at the end of November, the lowest since May, compared to 144 kt at the end of September.

Class I nickel metal stocks remain high The growth in stainless melt output has been noted across nickel-containing austenitics (up by 76 kt to November 2018) as well as nickel-free ferritics (up by 50 kt). We expect annual growth of 3.5-4.5% in each when final year figures are released next month. We did notice a dip in the austenitic proportion for August and September, to below 50%, and this probably contributed to the increasing stock of Class I nickel metal recorded for Japan since July. METI noted that metal inventories across producers, distributors and consumers had risen to 22.2 kt by September compared to 15.8 kt at the start of the year and 17.3 kt in July. The austenitic portion of the product mix edged back up again during October and November, to about 52%, but total output for those two months remained at lower levels of approximately 262 kt a month compared to 277 kt a month from January. This also helped to keep Class I nickel stocks at the relatively elevated levels of 22 kt.

Australian briquette imports have doubled Although the changes in stainless steel product mix have contributed to higher nickel stocks since the third quarter, a more direct explanation for the larger inventory comes from trade data. Japanese exports of Class I nickel to November were down by 30% on full year 2017, from 23.5 kt to 16.5 kt, with the biggest reductions tied to China, India and the UK. Meanwhile, Japanese imports of Class I nickel increased by 50% from 33.7 kt in 2017 to 49.4 kt last year. Briquettes from Madagascar (up to 12.6 kt from 9.7 kt in 2017) and cathode from Norilsk (up to 2.6 kt from 0.6 kt) were beneficiaries. But it is briquettes from the Australian producers of Murrin Murrin and Kwinana that took the main share, consistently exceeding 1.5 kt a month since January, compared with 600 t a month in 2017.

Demand from battery segment strengthening In our view, we believe that the greater imports of briquettes, along with similarly large purchases of nickel sulphate and powder, have been earmarked for the chemicals markets of plating and lithium ion batteries. Imports of sulphate rose by 12.7% in 2018, at 67.6 kt (gross), compared with 56 kt for 2017, with producers in Taiwan accounting for 72% of total receipts. Meanwhile, powder imports were also up by 10%, to 8.2 kt in 2018. For both products, Japanese exports were also lower in 2018 compared to the prior year. The effect of the net trade across all of these nickel products has contributed to a stock build as well as a higher level of consumption. Ultimately, most of this net additional product is being converted to precursors for electric vehicle battery cells, and we reflect this in our estimate of nickel demand from this segment of 28 kt in 2018 compared with 22 kt in 2017. If accurate, our forecast for 2019 of 31 kt might have to be moved up.

China

Chinese nickel demand in 2018 revised down We have revised up Chinese stainless melt production in 2018 by 200 kt to 26.2 Mt based on the latest available data. At the same time we have increased scrap use at some mills, especially Tsingshan FuAn, which reduced our estimate of primary nickel uptake. The net change is a cut in Chinese nickel demand of 20 kt, to 1.16 Mt in 2018.

Higher scrap ratio at Tsingshan in 2018 In Tsingshan's case, the rapid ramp up to 2 Mt of stainless production in Indonesia absorbed most of the NPI produced at its Morowali facility, which had previously been shipped back to China. This restricted the group's access to nickel units for its Chinese mills, thereby heightening the need for more scrap. Tsingshan allegedly profited from a surge in scrap availability in Q3 2018, due to old plant shutdowns in the Dainan region of Fujian province, not far from FuAn. It is unclear how much scrap use increased but we are not convinced by reports of scrap ratios at FuAn rising to 70-80% (compared to the usual level of 20-23%). Such high levels are without precedent and need further checking. However, assuming that the challenge of sourcing nickel units across the Tsingshan group of operations will not get any easier, especially once the third Indonesian stainless line comes on line, the prospect of scrap ratios in the FuAn and GuangQing mills increasing is likely.



Maintenance/holiday breaks in Q1 2019 China's stainless steel market is currently quiet due to the upcoming Chinese new year holiday. Stainless HR 304 prices picked up slightly, approaching CNY 14,000/t (US\$2,084/t), although this is still below the level needed to cover the production cost for standalone mills. With tight liquidity and a cautious sentiment in the market, stainless trading firms are circumspect about pre-holiday re-stocking, so orders to the mills are lower. This may be prompting more or longer maintenance breaks over the holiday period. In total, we have reduced our forecast of stainless melt output in 2019 by 235 kt to reflect cuts that had not already been allowed for. Not all companies shut down for the holiday, Tisco, Jisco, Delong and Baosteel Desheng among them. But others, such as Fuxin, Huale and Zhongjin are planning breaks of up to two weeks or more. Tsingshan may reduce output of 300 series by 20% in February.

Longer breaks at some mills Lisco has scheduled 20-30 days of maintenance that could reduce output by 100 kt, mostly 200 and 400 series. Henan Jinhui is another mill undertaking extended maintenance for three weeks, starting 25 January. While the cut of up to 50 kt will only be 200 series, the upgrade will then enable Jinhui to produce 300 series for the first time. After reducing output by 20% in January, Behei Chengde will cut production by a further 40% in February, amounting to a total cut of 100-150 kt, evenly distributed between 200 and 300 series. The company intermittently produced small quantities of 400 series in 2018 but there is no plan to produce any of this grade before March this year.

Europe

Stainless contraction in 2019 Growth in European stainless steel production slowed down in 2018, increasing by less than 1%, to 7.54 Mt, compared with 1.7% in 2017. Actually, year on year production growth was maintained at around 1.6% through the first three quarters of 2018 but activity dipped in Q4, with output falling year on year by 2.1% in those three months. We expect the downward trend to continue into 2019 not least because of mounting uncertainty surrounding Brexit. We currently forecast a 5.5% decline over the whole year, to 7.13 Mt. This would be followed by recovery in 2020, to 7.47 Mt. Despite the increase in stainless melt output in 2018, uptake of primary nickel decreased by 7%, to 164 kt, due to a rise in scrap ratios facilitated by higher nickel prices. Primary nickel use in 2019 will remain close to this level.

Mills need nickel prices to go up further Comments received alluding to an improvement in orders for stainless flat products will be encouraging to the mills but this is no more than would usually be expected for this time of year. The first half is normally the strongest business period of any year in Europe. However, while the higher late-January nickel prices may spur optimism, that improvement did not come in time to prevent a cut in alloy surcharges for February. This may restrain stainless orders for the time being. That delivery times are also currently short indicates that there is more than enough product availability at present. This too will restrain orders, as buyers can easily get hold of the stainless they need; there is no need for urgency. However, if further increases in the nickel price convince buyers that stainless prices will rebound in March, orders could surge in February before that rally, thereby tightening supply and giving the mills more freedom to raise melting rates.

North America

Flat start to the year in the USA Nickel demand in the USA got off to a marginally better start in 2019 than some traders were expecting just before the Christmas holidays, but that is not an endorsement of anticipated or ongoing strength. Stock replenishment, as factories returned to work in early January, was no more than expected. In light of the continued stainless scrap supply excess, with spot valuations for nickel in scrap at 65% of LME and heading lower, some traders are relieved that their pre-holiday fear of disaster for primary nickel offtake has not yet materialised. Perhaps this relatively upbeat sentiment has also carried through to price makers at the LME, where nickel exceeded US\$12,000/t (cash) on 30 January, a level last seen in late October.

Growth of 3% in stainless in 2019 "Relative" and "perspective" are likely to be key words for America's stainless and nickel markets this year, as whatever growth takes place over the year will not be as good as the average of 6.6% a year since 2016. In our opinion, given the layers of doubt at the macro-economic level in the USA (see below), our forecast of 3% year on year growth in stainless melt output should be viewed as a good result.

ATI still at disadvantage to NAS and Outokumpu Our outlook for 3% growth in 2019 depends on the Allegheny (ATI) slab import decision. As ATI has still not received a tax exemption to import slab from Indonesia, we expect its domestic competitors



to continue melting at high capacity utilisation rates, at least for the first half of the year. Consequently, we have maintained our bias towards a stronger H1 for national melt production (1.49 Mt) than H2 (1.44 Mt).

Government shutdown hinders 232 exclusion process As the Section 232 review process was held up during the recent 35-day long partial government shutdown, with no exemptions granted since 21 December, ATI's relief at seeing the end to the impasse in late-January might only last until 15 February, when the two parties in Congress clash once again. As heavy a toll as another shutdown would take on the economy, following the US\$3 billion cost from the first, attention will turn quickly to signs of progress in trade dispute between the USA and China. The deadline for success is 2 March, if the US is to refrain from imposing a tariff hike from 10% to 25% on a further US\$200 billion worth of imports from China.

Macro economics weigh heavily in 2019 The main effect of this unresolved friction is a lack of visibility for corporations trying to plan production rates, and uncertainty among consumers who are vulnerable to rising costs. From trade tariffs to higher interest on mortgages and auto loans, owing to the Federal Reserves policies to stem inflation, the threats to consumer-led growth loom large. The American Automotive Policy Council said this month that vehicle sales will suffer in 2019, as tariffs add as much as US\$300 a unit to the cost of a new vehicle. Yet, the Fed promises further rate hikes to the 0.25-0.41% already applied on auto loans during Q4 2018.



Global quarterly refined consumption (kt)

	2018	2019					2020				
	Yr	Q1	Q2	Q3	Q4	Yr	Q1	Q2	Q3	Q4	Yr
Africa	31	8	8	7	8	30	8	8	7	7	29
RSA	28	7	7	7	7	28	7	7	6	7	27
Other	2	1	1	1	1	2	1	1	1	1	2
Asia	1731	428	453	459	475	1816	475	493	496	500	1963
China	1160	287	309	311	319	1226	313	325	326	327	1291
India	57	15	16	15	16	62	16	17	16	18	67
Indonesia	156	38	38	43	49	168	55	61	63	64	242
Japan	164	41	41	41	42	165	42	40	41	42	166
Korea	100	25	26	25	25	100	24	24	24	24	97
Taiwan	57	14	14	15	15	58	15	15	16	16	62
Other	37	9	9	9	9	38	10	10	10	10	39
Europe	336	84	83	77	83	328	89	90	82	86	346
Austria	10	3	3	2	2	10	3	3	2	3	10
Belgium	40	11	10	10	10	41	11	12	10	11	44
Czech Rep	4	1	1	1	1	4	1	1	1	1	5
Finland	31	8	8	8	8	32	8	8	8	8	33
France	27	7	6	6	6	26	7	7	6	7	27
Germany	69	17	17	16	17	67	18	18	17	18	71
Italy	54	14	14	12	13	52	14	14	12	13	54
Poland	4	1	1	1	1	4	1	1	1	1	4
Romania	1	0	0	0	0	1	0	0	0	0	1
Slovenia	2	1	1	1	1	2	1	1	1	1	2
Spain	33	8	8	8	8	33	9	9	8	9	35
Sweden	28	6	6	6	6	24	7	7	6	7	27
UK	18	5	5	4	5	18	5	5	5	5	20
Ukraine	7	2	2	2	2	7	2	2	2	2	7
Other	7	2	2	2	2	7	2	2	2	2	7
Latin America & Caribbean	17	4	4	5	4	17	4	5	5	5	19
Brazil	15	4	4	4	4	15	4	4	4	4	17
Cuba	1	0	0	0	0	1	0	0	0	0	1
Other	1	0	0	0	0	1	0	0	0	0	1
North America	151	39	39	36	39	153	39	40	36	39	155
Canada	7	2	2	2	2	8	2	2	2	2	8
Mexico	3	1	1	1	1	4	1	1	1	1	4
USA	140	36	36	33	36	142	37	37	34	36	143
Oceania	5	1	2	3	4	11	5	5	6	6	22
Australia	4	1	2	3	4	10	5	5	6	6	21
Other	1	0	0	0	0	1	0	0	0	0	1
Russia & Caspian	33	8	9	8	8	33	9	9	9	8	34
Russia	33	8	9	8	8	33	9	9	9	8	34
Global	2303	573	598	595	622	2388	628	649	639	653	2569

Source: Wood Mackenzie



Quarterly global stainless steel production (kt raw steel basis)

	2018	2019					2020				
	Yr	Q1	Q2	Q3	Q4	Yr	Q1	Q2	Q3	Q4	Yr
Africa	581	145	140	130	145	560	160	155	135	150	600
RSA	581	145	140	130	145	560	160	155	135	150	600
Asia	39024	9324	10121	10209	10643	40297	10575	11133	11052	11234	43994
China	26200	6140	6812	7029	7219	27200	7064	7432	7492	7512	29500
India	3750	960	1080	920	1040	4000	1056	1188	1012	1144	4400
Indonesia	2203	540	540	620	700	2400	780	860	890	900	3430
Japan	3295	840	815	770	790	3215	797	765	760	780	3102
Korea	2446	584	614	600	614	2412	598	598	598	598	2392
Taiwan	1130	260	260	270	280	1070	280	290	300	300	1170
Other	0	0	0	0	0	0	0	0	0	0	0
Europe	7542	1905	1833	1610	1782	7130	1973	1995	1670	1834	7473
Austria	79	19	19	16	17	71	19	18	17	18	73
Belgium	1656	420	400	350	380	1550	433	433	373	413	1653
Czech Rep	8	2	2	2	2	7	2	2	2	2	8
Finland	1348	330	310	310	330	1280	350	350	335	306	1341
France	310	79	70	60	70	279	80	80	60	70	290
Germany	440	111	110	80	106	406	115	120	80	105	420
Italy	1496	380	375	300	350	1405	381	391	311	361	1445
Poland	2	0	0	0	0	1	0	0	0	0	2
Slovenia	143	36	33	37	37	143	37	39	37	37	150
Spain	1006	260	250	235	250	995	273	273	231	259	1035
Sweden	632	160	160	131	145	596	176	176	131	161	643
UK	311	80	75	60	66	281	77	83	63	72	295
Ukraine	111	29	30	29	30	116	30	30	30	29	118
Latin America & Caribbean	420	93	102	111	102	408	100	110	114	102	426
Brazil	412	91	100	109	100	400	98	108	112	100	418
Cuba	8	2	2	2	2	8	2	2	2	2	8
North America	2908	750	760	733	720	2964	755	736	733	730	2954
Canada	38	10	10	8	10	39	10	11	8	10	39
USA	2870	740	750	725	710	2925	745	725	725	720	2915
Russia & Caspian	199	47	49	44	44	184	52	55	50	49	206
Russia	199	47	49	44	44	184	52	55	50	49	206
Global	50675	12264	13005	12837	13436	51543	13615	14184	13754	14099	55653

Source: Wood Mackenzie



Global refined nickel consumption (kt)

	2015	2016	2017	2018	2019	2020	2021
Africa:	23	27	29	31	30	29	30
RSA	21	25	26	28	28	27	27
Other	2	2	2	2	2	2	2
Asia:	1336	1515	1614	1731	1816	1963	1994
China	999	1128	1160	1160	1226	1291	1300
India	33	47	59	57	62	67	70
Indonesia	0	2	52	156	168	242	254
Japan	142	151	153	164	165	166	165
Korea	80	91	90	100	100	97	101
Taiwan	48	62	65	57	58	62	64
Other	34	35	36	37	38	39	40
Europe:	321	345	348	336	328	346	348
Austria	9	10	10	10	10	10	10
Belgium	39	42	41	40	41	44	45
Czech Rep	4	4	5	4	4	5	5
Finland	31	40	36	31	32	33	33
France	27	27	27	27	26	27	27
Germany	66	65	70	69	67	71	71
Italy	52	55	55	54	52	54	54
Poland	4	4	4	4	4	4	4
Romania	1	1	1	1	1	1	1
Slovenia	3	3	3	2	2	2	3
Spain	33	35	35	33	33	35	36
Sweden	21	28	29	28	24	27	27
UK	18	19	19	18	18	20	20
Ukraine	6	6	7	7	7	7	7
Other	6	6	6	7	7	7	7
Latin America & Caribbean	18	17	14	17	17	19	19
Brazil	16	15	12	15	15	17	16
Cuba	1	1	1	1	1	1	1
Other	1	1	1	1	1	1	2
North America:	137	130	148	151	153	155	156
Canada	6	7	7	7	8	8	8
Mexico	4	4	4	3	4	4	4
USA	127	120	137	140	142	143	145
Oceania:	4	4	4	5	11	22	25
Australia	4	4	4	4	10	21	24
Other	1	1	1	1	1	1	1
Russia & Caspian:	30	31	32	33	33	34	35
Russia	30	31	32	33	33	34	35
Global:	1869	2070	2189	2303	2388	2569	2607
	1.5%	10.7%	5.8%	5.2%	3.7%	7.6%	1.5%

Source: Wood Mackenzie



Global stainless steel production (kt raw steel basis)

	2015	2016	2017	2018	2019	2020	2021
Africa:	515	582	591	581	560	600	605
RSA	515	582	591	581	560	600	605
Asia:	31698	34479	36799	39024	40297	43994	44979
China	22000	24500	25660	26200	27200	29500	30262
India	3300	3324	3486	3750	4000	4400	4500
Indonesia	0	25	703	2203	2400	3430	3593
Japan	3062	3093	3168	3295	3215	3102	3040
Korea	2224	2273	2406	2446	2412	2392	2385
Taiwan	1109	1263	1376	1130	1070	1170	1199
Other	3	0	0	0	0	0	0
Europe:	7271	7369	7493	7542	7130	7473	7528
Austria	66	67	71	79	71	73	72
Belgium	1541	1605	1627	1656	1550	1653	1661
Czech Rep	8	7	7	8	7	8	8
Finland	1344	1409	1372	1348	1280	1341	1349
France	291	288	293	310	279	290	292
Germany	459	414	436	440	406	420	422
Italy	1452	1420	1470	1496	1405	1445	1438
Poland	1	2	2	2	1	2	2
Slovenia	156	149	147	143	143	150	155
Spain	979	986	1003	1006	995	1035	1063
Sweden	621	632	639	632	596	643	656
UK	247	283	312	311	281	295	293
Ukraine	105	109	115	111	116	118	119
Latin America & Caribbean	425	445	438	420	408	426	427
Brazil	417	437	430	412	400	418	419
Cuba	8	8	8	8	8	8	8
North America:	2370	2511	2789	2908	2964	2954	2887
Canada	25	30	35	38	39	39	40
USA	2345	2481	2754	2870	2925	2915	2847
Russia & Caspian:	172	181	177	199	184	206	217
Russia	172	181	177	199	184	206	217
Global:	42452	45567	48287	50675	51543	55653	56643
	0.1%	7.3%	6.0%	4.9%	1.7%	8.0%	1.8%

Source: Wood Mackenzie

Supply

Refined nickel up 7% to 2.41 Mt in 2019 Refined nickel production grew by 5.5% in 2018 to 2.24 Mt. We forecast a further increase of 7.3% in 2019 to 2.41 Mt and continued growth through to 2021, reaching 2.6 Mt by that time. A large proportion of this increase is due to continued growth in Indonesian and Chinese NPI output.

Mined nickel production to reach 2.5 Mt in 2019 As with refined output, mined nickel production rose to 2.28 Mt in 2018. Further growth in 2019 will increase output to 2.5 Mt, and then 2.6 Mt by 2021.

Indonesian blast furnaces suspended Rising coke prices have resulted in production suspensions at several blast furnace NPI plants in Indonesia. PT Central Omega ceased production in January. Zhenshi Gebe (PT Fajar Bhakti) continued to operate



despite operating losses. Indications are that the company will continue to operate until new RKEF lines are constructed at Weda Bay, but there is clearly a possibility that it could be placed on care and maintenance if there are no improvements in nickel price or falls in coke price. Reports also suggest that Bintang ceased production this month. While we have made downward adjustments in output for these operations, we currently assume that they will restart at some point later in the year. Clearly, however, there is a risk that further reductions in forecast production may well be necessary if those re-start do not take place.

PT Hengjaya to start ahead of schedule Nickel Mines Limited, in partnership with Tsingshan at the Morowali Industrial Park (IMIP), started building two 42 MVA RKEF lines in mid- 2018, with first production anticipated by the June quarter of 2019. Commissioning of these first two line is scheduled for Q1, with 80% of capacity expected to be reached within 6-8 weeks from commissioning. Nickel Mines is also participating in the construction of two further lines (known as Ranger Nickel), which are projected to be operational before the end of the year. In total, the four lines will add approximately 33 ktpa nickel in NPI to the existing 20 lines (180 ktpa capacity). Two more lines are expected to added by Tsingshan, possibly in partnership with the Zhenshi Group, taking the total installed NPI capacity in IMIP to around 230 ktpa.

Most of Tsingshan's NPI will be used on site in stainless Tsingshan consumed around 156 kt nickel in the production of 2.2 Mt of stainless steel at IMIP during 2018. When the third 1 Mt stainless line is put into production, the annualised nickel requirement will rise to around 210 ktpa. Our forecast nickel production at IMIP and the timing of the third stainless line coming on-stream, suggest that exports of NPI from IMIP into China are likely to continue, probably to support Tsingshan's stainless facilities in FuAn and GuangQing.

QMB project on stream by 2021 Following the bold announcements late last year that the new HPAL facility being built at IMIP by joint venture partners Tsingshan, GEM, Brunp and Hanwa, would be on-stream in 2019, it now seems that 2021 is a more realistic start date. Recent reports indicate that GEM is talking with Sumitomo Mitsui Banking Corporation and the Bank of China regarding funding for the US\$700 million development. The announcement indicated the company hoped to have financing in place by May this year and construction would be complete by end-December 2020.

QMB will make NiSO₄ and Ni(OH)₂ While there has been some confusion surrounding what nickel product would be produced at the Indonesian facility (initial reports implying an MHP intermediate that would be processed in China), our current view is that the final products will be nickel sulphate and nickel hydroxide (Ni(OH)₂). Total plant capacity will be 50 ktpa nickel (plus 4 ktpa cobalt), with final output being up to 150 ktpa NiSO₄ (ca. 33 ktpa nickel) and up to 50 ktpa of nickel hydroxide (ca. 32 ktpa nickel on a dry basis). The former would be for use in ternary battery manufacture and the latter for NiMH batteries. The mix of output would depend on market requirements.

Ramu “integrated refinery” enters production Metallurgical Company of China (MCC) commissioned the first stage of its nickel sulphate production facility in Caofeidian in late December 2018. We understand that the initial capacity is for 50 ktpa NiSO₄ and feed will be in the form of a mixed hydroxide from Ramu in PNG, in which MCC is a major shareholder. The company noted that products had not yet been sold to the market while it “fine tunes” the quality of the material. MCC is also currently building a ternary precursor plant in Caofeidian. This is a two stage development, the first costing 2.34 billion yuan (ca. US\$350 million) to produce 40 ktpa of NCM 611 precursors. The second phase, costing 1.34 billion yuan (ca. US\$200 million) will produce 60 ktpa of NCM 811 precursors. The nickel sulphate product will be sold on the open market until the precursor plants are ready. While the initial capacity of the Caofeidian refinery is around 10 ktpa nickel, it appears likely that this will be raised to take more of the feed from Ramu in the coming years.

Shandong Xinhai ramps up NPI in China Shandong Xinhai Technology is the largest producer of NPI in China. Our figures indicate that the company produced around 20% of all domestic NPI in 2018. The company was operating 25 33 MVA lines and is adding capacity through its Phase III expansion. Phase III consists of eight 48 MVA furnaces, the first of which commenced operating in January. While this was two months later than originally planned, the company has indicated that it hopes to have all eight lines operating by the end of the first half of the year. This would increase capacity by around 1.2 Mtpa gross (ca. 100 ktpa nickel), taking the total gross capacity to 3 Mtpa (ca. 250-260 ktpa nickel). Our production outlook is conservative, reaching 135 kt nickel in 2020, on the assumption that further increases in production will be dependent upon the company completing its proposed 1.2 Mtpa stainless steel plant.



Production cut at Doniambo due to strikes On 3 January the Thio and Tiebaghi mines, owned by SLN in New Caledonia, ceased production due to strike action. These were joined by the Nepoui site in the second week of January and at present no resolution has been found for mining to re-start. The strike action has been undertaken because unions oppose changes to work patterns and hours. These were proposed as part of the restructuring of SLN to reduce operating costs and improve competitiveness. SLN's fourth mine, Kouaoua, continues to operate under the new work regime. The strikes have resulted in ore stocks being run down at Doniambo (reported to be at only a third of usual levels), which has led to the plant having to reduce FeNi production. We have lowered output at Doniambo to 53 kt (contained) in 2019, but without resolution to the strike action more significant reductions may be necessary in the future.

Larco to cut output by 20% to “settle debt” An agreement reached in mid-January between the Greek government and Larco SA will see Larco cut nickel production by 20% and pay €4 million (US\$4.57 million) per month to the Public Power Company (PPC) for electricity consumption. PPC had threatened to stop supplying Larco with electricity if no deal could be reached. Larco has outstanding debts of around €280 million (US\$320 million) to PPC for prior years electricity consumption.



Global nickel mine production/capability (kt)

	2015	2016	2017	2018	2019	2020	2021
Botswana	24	11	0	0	0	0	0
Ivory Coast	0	0	0	6	6	6	6
Madagascar	55	49	42	39	45	45	42
South Africa	53	51	48	48	48	48	48
Zambia	0	0	0	0	3	5	5
Zimbabwe	11	13	12	11	11	12	7
Total Africa	142	124	102	105	113	115	108
China	98	94	95	95	95	97	97
Indonesia	152	215	417	612	732	755	825
Myanmar	19	19	21	24	19	25	25
Philippines	483	351	371	327	359	361	364
Vietnam	9	4	0	0	0	0	0
Total Asia	760	682	905	1058	1205	1238	1311
Albania	2	0	1	1	1	1	1
Finland	21	42	61	60	58	58	59
Greece	22	22	22	21	18	18	18
Kosovo	4	3	4	3	4	6	6
Norway	0	0	0	0	0	0	0
Republic of Macedonia	3	3	5	5	5	5	5
Spain	7	0	0	0	0	0	0
Turkey	2	2	4	4	6	6	6
Ukraine	0	0	3	0	4	0	0
Total Europe	62	73	100	94	96	94	95
Brazil	117	95	69	46	78	76	76
Colombia	41	41	45	48	43	40	36
Cuba	56	53	54	54	55	55	57
Dominican Republic	0	7	16	16	16	16	16
Guatemala	47	44	40	40	52	62	66
Venezuela	5	0	0	0	0	0	0
Total Latin America & Caribbean	266	241	224	204	245	250	251
Canada	234	225	209	185	188	185	192
USA	27	24	22	18	14	18	15
Total North America	261	249	231	203	202	203	206
Australia	188	153	158	157	164	146	120
New Caledonia	184	214	215	214	220	236	242
Papua New Guinea	30	26	39	40	40	40	40
Total Oceania	402	393	412	411	425	422	402
Kazakhstan	0	0	1	1	1	1	1
Russian Federation	214	192	208	206	208	212	215
Total Russia & Caspian	214	192	209	206	208	212	216
Total World	2107	1953	2183	2282	2494	2534	2588
<i>Disruption allowance + highly prob proj</i>				0	-39	53	139
Forecast mine production	2107	1953	2183	2282	2455	2587	2727
	3.2%	-7.3%	11.8%	4.5%	7.6%	5.4%	5.4%

Source: Wood Mackenzie



Global nickel smelter production/capability (kt)

	2015	2016	2017	2018	2019	2020	2021
Botswana	17	18	0	0	0	0	0
Madagascar	48	43	36	34	39	39	37
South Africa	39	38	38	38	38	38	38
Zimbabwe	4	5	5	5	5	5	5
Total Africa	108	104	80	77	81	81	79
China	546	498	532	603	628	628	631
Indonesia	129	197	289	370	457	528	589
Japan	63	53	48	50	53	53	53
Myanmar	18	18	20	23	18	24	24
Philippines	49	46	51	51	54	56	58
South Korea	39	46	47	48	50	50	50
Total Asia	844	858	988	1145	1260	1339	1405
Austria	1	1	1	1	1	1	1
Finland	28	41	46	59	60	55	55
Greece	17	17	17	16	14	14	14
Kosovo	8	3	8	5	7	8	8
Macedonia	18	11	2	12	16	20	20
Turkey	0	1	2	5	5	5	5
Ukraine	18	18	15	16	17	18	18
Total Europe	90	91	91	113	120	121	121
Brazil	73	75	69	65	70	69	69
Colombia	37	37	41	43	39	36	32
Cuba	48	45	45	46	47	47	48
Dominican Republic	0	7	15	15	15	15	15
Guatemala	11	9	12	17	17	18	22
Venezuela	4	0	0	0	0	0	0
Total Latin America & Caribbean	172	173	182	185	188	185	185
Canada	216	248	224	185	193	194	194
Total North America	216	248	224	185	193	194	194
Australia	166	154	143	126	132	122	122
New Caledonia	93	105	112	115	120	134	140
Papua New Guinea	26	22	35	35	35	35	35
Total Oceania	285	281	290	276	287	292	298
Russian Federation	238	210	215	212	210	209	209
Total Russia and the Caspian	238	210	215	212	210	209	209
Total World	1952	1965	2069	2192	2340	2421	2492
<i>Disruption allowance + highly prob proj</i>				0	-45	-36	-16
Forecast smelter production	1952	1965	2069	2192	2295	2385	2476
	-3.5%	0.6%	5.3%	6.0%	4.7%	3.9%	3.8%

Source: Wood Mackenzie



Global finished nickel production/capability (kt)

	2015	2016	2017	2018	2019	2020	2021
Madagascar	47	42	35	33	38	38	36
South Africa	42	42	43	42	42	42	42
Zimbabwe	0	0	0	0	0	0	0
Total Africa	89	84	78	75	80	80	78
China	630	594	661	713	767	764	783
India	3	2	2	4	3	3	3
Indonesia	47	119	212	293	370	440	493
Japan	194	192	177	166	173	177	181
Philippines	0	0	0	0	0	0	0
Myanmar	18	18	20	23	18	24	24
South Korea	40	47	50	51	54	54	54
Total Asia	932	973	1123	1251	1384	1462	1538
Austria	1	1	1	1	1	1	1
Belgium	7	8	9	9	10	10	10
Finland	43	54	60	60	63	63	78
France	7	5	2	3	6	8	10
Germany	7	8	8	9	9	10	10
Greece	17	17	17	16	14	14	14
Kosovo	8	3	8	5	7	8	8
Macedonia	18	11	2	12	16	20	20
Norway	91	93	87	90	92	92	92
Spain	0	0	1	1	1	1	1
UK	39	43	37	38	41	40	39
Ukraine	18	18	15	16	17	18	18
Total Europe	257	260	246	260	276	284	300
Brazil	79	77	69	65	70	69	69
Colombia	37	37	41	43	39	36	32
Cuba	14	13	14	15	15	15	15
Dominican Republic	0	7	15	15	15	15	15
Guatemala	11	9	12	17	17	18	22
Venezuela	4	0	0	0	0	0	0
Total Latin America & Caribbean	144	142	150	154	156	153	152
Canada	150	161	151	129	133	135	137
USA	0	0	0	0	0	0	0
Total North America	150	161	151	129	133	135	137
Australia	132	119	112	114	112	112	112
New Caledonia	77	93	105	108	117	131	137
Total Oceania	209	212	217	222	229	244	250
Russian Federation	234	195	161	153	149	147	147
Total Russia & Caspian	234	195	161	153	149	147	147
Total World	2016	2027	2126	2243	2407	2504	2602
<i>Disruption allowance + highly prob proj</i>				0	-47	-38	-20
Forecast refinery production	2016	2027	2126	2243	2360	2466	2582
	-0.5%	0.6%	4.9%	5.5%	5.2%	4.5%	4.7%

Source: Wood Mackenzie



Nickel stocks and premiums

	2018	2018	2018	2018					2019
		Q3	Q4	Aug	Sep	Oct	Nov	Dec	Jan
Stocks - kt									
Antwerp	0.3	0.3	0.3	0.31	0.3	0.3	0.3	0.3	0.3
Baltimore	2.3	2.5	2.3	2.59	2.5	2.4	2.3	2.3	2.1
Busan	0.8	0.5	0.8	0.59	0.5	1.2	1.0	0.8	0.6
Dubai	13.7	12.9	13.7	12.91	12.9	12.9	12.4	13.7	11.1
Gwangyang	1.2	1.3	1.2	1.43	1.3	1.3	1.2	1.2	0.6
Johor	80.1	94.3	80.1	96.26	94.3	87.7	83.9	80.1	79.0
Kaohsiung	30.8	33.6	30.8	35.89	33.6	31.7	31.2	30.8	30.4
Liverpool	0.3	0.3	0.3	0.30	0.3	0.3	0.3	0.3	0.3
Moerdijk	2.6	2.8	2.6	2.81	2.8	2.8	2.7	2.6	2.6
Port Klang	0.5	0.4	0.5	0.53	0.4	0.3	0.5	0.5	1.3
Rotterdam	38.4	42.3	38.4	45.46	42.3	41.3	40.2	38.4	36.3
Singapore (City)	35.2	37.0	35.2	39.34	37.0	36.6	36.6	35.2	37.4
Total LME Stocks	206.3	228.1	206.3	238.4	228.1	218.8	212.8	206.3	202.0
Cancelled warrants	55.3	49.9	55.3	56.8	49.9	47.9	50.0	55.3	59.7
Shanghai Futures Exchange	15.3	17.0	15.3	17.7	17.0	14.4	15.3	15.3	12.7
Prices									
LME									
Cash (\$/t)	13118	13263	11512	13429	12524	12323	11249	10833	11366
3-month (\$/t)	13182	13354	11593	13516	12616	12408	11325	10917	11426
Spread (3-month - cash)	64	91	81	87	92	84	76	84	60
SHFE									
Front month settlement Yuan/t	103020	108525	95343	110528	103042	103706	93686	89641	92230
3-month settlement Yuan/t	103562	108899	95176	111077	103346	102712	93780	89928	92613
Front month settlement \$/t	13326	13635	11774	13793	12849	12780	11540	11126	11552
3-month settlement \$/t	13398	13682	11753	13862	12886	12657	11551	11161	11597
Physical Premiums									
W. Europe Plating	515	408	467	403	420	420	511	515	515
W. Europe Melting	238	282	243	290	265	245	240	238	238
Russian Uncut	230	245	235	250	235	235	235	230	225
US Melting Premium	600	700	687	700	700	700	682	600	553
US Plating Premium	950	950	950	950	950	950	950	950	880
Exchange Rates/US\$									
Australian \$	1.340	1.367	1.395	1.365	1.388	1.408	1.379	1.397	1.401
Canadian \$	1.289	1.307	1.321	1.304	1.304	1.301	1.320	1.345	1.333
Yen	110.54	111.47	112.74	111.00	112.03	112.75	113.33	112.11	108.91
Yuan	6.601	6.805	6.917	6.849	6.854	6.924	6.939	6.886	6.801
Euro	0.754	0.860	0.876	0.866	0.857	0.871	0.880	0.879	0.876

Source: Thomson Reuters Datastream; Wood Mackenzie



Global refined nickel supply-demand balance (kt)

	2015	2016	2017	2018	2019	2020	2021
Mine Production/Capability	2107	1953	2183	2282	2494	2534	2588
Highly probable projects					4	8	8
Forecast disruption allowance					-50	-51	-52
Base case mine output	2107	1953	2183	2282	2448	2491	2544
Probable mine projects					7	97	183
Additional adjustment							
Forecast market adjustment					7	97	183
Actual/forecast mine output	2107	1953	2183	2282	2455	2587	2727
Smelter Production Capability	1952	1965	2069	2192	2340	2421	2492
Highly probable projects							
Forecast disruption allowance					-47	-48	-50
Base case smelter output	1952	1965	2069	2192	2293	2373	2442
Probable smelter projects					2	13	34
Additional adjustment							
Forecast market adjustment					2	13	34
Actual/forecast smelter output	1952	1965	2069	2192	2295	2385	2476
Refined Production Capability	2016	2027	2126	2243	2407	2504	2602
Highly probable projects							
Forecast disruption allowance					-48	-50	-52
Base case refined output	2016	2027	2126	2243	2359	2454	2550
Probable refinery projects					2	12	32
Additional adjustment							
Forecast market adjustment					2	12	32
Actual/forecast refined output	2016	2027	2126	2243	2360	2466	2582
DLA/ Jap. govt sales							
Primary Nickel Supplies	2016	2027	2126	2243	2360	2466	2582
Consumption	1869	2070	2189	2303	2388	2569	2607
Balance	147	-43	-63	-59	-28	-103	-25
Estimated stocks	1510	1467	1404	1344	1316	1214	1188
- in days of Consumption	295	259	234	213	201	173	166
LME Price (US c/lb)							
Current c/lb	537	435	472	595	550	695	650
2019 \$	582	466	495	608	550	682	625
LME Price (US \$/t)							
Current \$/t	11832	9594	10403	13114	12119	15331	14339
2019 \$	12828	10273	10905	13408	12119	15030	13782

Source: Wood Mackenzie

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